

The University of the South Pacific

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School of Information Technology, Engineering, Mathematics and Physics

CS324: Distributed System

Test 1 – Semester 2, 2022

Learning Outcome: Outline the importance of distributed systems over stand-alone systems

Time Allowed: 50 mins

Total Marks: 30 (10% of final grade)

Name: _____

Student ID: _____

INSTRUCTIONS:

- All questions are compulsory.
- All answers must be written in the spaces provided in this booklet.
- You are allowed to use a silent, non-programmable calculator.
- Show all working.

1. Explain why middleware is the OS of the distributed systems. (3 marks)

ANS:

In a sense, middleware is the same to a distributed system as what an operating system is to a computer: a manager of resources offering its applications to efficiently share and deploy those resources across a network.

Next to resource management, it offers services that can also be found in most operating systems, including:

- Facilities for interapplication communication.
- Security services.
- Accounting services.
- Masking of and recovery from failures.

2. Discuss the two characteristics of the distributed systems. (4 marks)

ANS:

1. Collection of autonomous computing elements
2. Single coherent systems

3. Name 2 techniques to aid scalability of distributed systems. Provide an example for each. (4 marks)

Ans:

1. Replication: For the Web, replicate copies of the hosting Web server objects and pages to reduce latency and reduce server load. Akamai (and other CDN's) are examples.
2. Distribution: dividing up load among many different nodes, typically in a hierarchical fashion, increasing scalability over a single, centralized server. Name resolution on the Internet, the Domain Name Service (DNS), is an example.

4. Give an example of a Client-Server system with a "fat" client. What would be the effect of making the client "thin" for this system? (3 marks)

Answer: A Web browser is a "fat" client since most of the computation takes place at the end-user computer. If a Web browser was made "thin" then basic commands, such as data entry, mouse click information, etc. would be transmitted to the server. The server would then have to determine the event, say if a link was clicked, parse the corresponding HTML and send an updated picture of the Web page to the client. It is quite likely latency, in particular, would be more noticeable. In addition, the Web server would likely become a bottleneck to performance. (There are other possible examples.)

5. Discuss the importance of distributed systems over standalone systems (3 marks).

Distributed systems **provide scalability and improved performance in ways that monolithic systems can't**, and because they can draw on the capabilities of other computing devices and processes, distributed systems can offer features that would be difficult or impossible to develop on a single system.

6. Explain the importance of threads in distributed systems? (3 marks)

ANS: it allows for building well-structured efficient servers, by having a simple mechanism handle concurrent requests

7. Why is it sometimes so hard to hide the occurrence and recovery from failures in a distributed system? (3 marks)

It is generally impossible to detect whether a server is actually down, or that it is simply slow in responding. Consequently, a system may have to report that a service is not available, although, in fact, the server is just slow.

8. What is an open distributed system and what benefits does openness provide? (3 marks)

An open distributed system offers services according to clearly defined rules. An open system is capable of easily interoperating with other open systems but also allows applications to be easily ported between different implementations of the same system.

9. Explain how does peer-to-peer system solve administrative scalability problem? (3 marks)

ANS:

The key to the answer lies in the fact that in many p2p systems end users are in control. As a result, there simply is not much formal administration to deal with, let alone administrative problems that are tightly coupled to organizational issues.

10. In some subject-based publish/subscribe systems, secure solutions are sought in end-to-end encryption between publishers and subscribers. However, this approach may violate the initial design goals of coordination-based systems. Explain how? (3 marks)

Ans: The problem is that end-to-end encryption requires that a publisher and subscriber know each other. This is against the initial design goal that processes should be referentially decoupled.